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Accurate Determination of Molecular Sizes of a Solute in Water From its Translational Self-Diffusion Coefficient

Francesco Zaccaria,*[a, b] Alceo Macchioni,[a] and Cristiano Zuccaccia*[a]

Determining accurate molecular dimensions in water, from measured translational self-diffusion coefficients (D_t), is extremely important in biochemistry, supramolecular chemistry, organometallic chemistry and beyond, but it still represents a big challenge especially for small and medium-sized molecules. Indeed, current semiempirical adaptations of the Stokes-Einstein equation, which allow accurate determination of molecular size of solutes in organic solvents, proved inadequate for aqueous systems. To overcome such a major limitation, herein, we introduce a novel approach that unlocks the quantitative interpretation of D_t in water. By analyzing ~70 diverse molecules with volumes ranging from 10^1 Å^3 to 10^3 Å^3, and selecting the partial molar radius (r_M) as a reliable proxy for the hydrodynamic radius (r_H), we derived a semiempirical equation that enables accurate determination of hydrodynamic volume (V_H) of solutes in aqueous solutions, effectively accounting for the distinctive hydrogen-bonding properties of water. This approach fills a crucial gap, enhancing precise molecular characterization of polar and non-polar solutes in water.

Introduction

Recent advancements in analytical methods,[1–3] particularly diffusion NMR spectroscopy (DOSY NMR),[4–5] have provided straightforward tools for accurately measuring translational self-diffusion coefficients (D_t) in solution. Quantitative analysis of D_t offers precious information for various applications, such as the characterization of complex mixtures,[6–7] determination of oligo-/polymerization degrees,[8] and analysis of intermolecular interactions.[9–10] The underlying requirement for all these applications is the determination of molecular size.[11–15]

The Stokes-Einstein (SE) equation is commonly employed to estimate molecular size by correlating the D_t of a solute with its hydrodynamic radius (r_H).[16] However, the original SE equation assumes idealized conditions (*i.e.* solutes having perfectly spherical shape and much larger dimensions than the solvent's), which are rarely met in real solutions.[11] To overcome these limitations, correction factors are introduced into a modified SE equation:

$$D_t = \frac{k_B T}{c f_s \pi \eta r_H} \tag{1}$$

where k_B is the Boltzmann constant, T the absolute temperature, η the fluid viscosity, and f_s and c are correction factors accounting for the shape and dimension of the solute, respectively.[11] The f_s parameters can be calculated based on the shape of the solute[17–20] and often approximated to 1 for small molecules (*i.e.* spherical shape; *vide infra*).[11–12] The c parameter is instead estimated using a semiempirical adaptation of Wirtz's microfriction theory equations,[21–22] as proposed by Chen and Chen:[23]

$$c = \frac{6}{1 + 0.695\left(\frac{r_{solv.}}{r_H}\right)^{2.234}} \tag{2}$$

where r_{solv} is the hydrodynamic radius of the solvent.

While this approach has been successful in various organic solvents,[12,20,24–26] it is inadequate for solutions in water, especially for small and medium-sized molecules. Given the centrality of aqueous systems in many fields such as biological and medicinal chemistry,[27–30] natural products,[31–34] supramolecular chemistry[35–39] and catalysis,[40–43] the absence of a reliable method for interpreting D_t in water is a significant limitation. By definition, hydrodynamic dimensions are inherently related to the nature of the solvent (*hydro*) and the *dynamics* of diffusion, which means they are influenced not only by the relative sizes of the solute and solvent but also by specific solute-solvent interactions.[44] The Chen and Chen equation (Equation (2)) explicitly accounts for relative dimensions through the r_{solv}/r_H ratio, while it implicitly addresses solute-solvent interactions through the semiempirical coefficients derived in organic solvents. Solute-solvent interactions can be assumed similar in the most common organic media, but they vary significantly in water, a structured solvent capable of forming strong hydrogen bonds with itself and with polar solutes.[45]

[a] *Department of Chemistry, Biology and Biotechnology and CIRCC, University of Perugia, Perugia, Italy*
[b] *Department of Chemical Science, Federico II University of Naples, Napoli, Italy*

Correspondence: *Francesco Zaccaria and Cristiano Zuccaccia, Department of Chemistry, Biology and Biotechnology and CIRCC, University of Perugia, Via Elce di sotto, 8-06123 Perugia, Italy.*
Email: francesco.zaccaria@unina.it and cristiano.zuccaccia@unipg.it

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Herein, we introduce an approach that incorporates the unique properties of water, enabling the quantitative interpretation of D_t in this ubiquitous solvent. By analyzing a broad and diverse set of known molecules, we developed a semiempirical $c = f\,(r_{solv}, r_H)$ expression, analogous to Equation (2), which allows accurate estimation of molecular size in water. This equation overcomes the limitations of previous methods, offering a robust and widely applicable tool for precise molecular characterization in aqueous systems.

Results and Discussion

Limitations of the Established Methods

To illustrate the limitations of existing methods, the SE equation with Chen and Chen's correction was applied to the well-known BF_4^- anion ($D_t = 16.0 \cdot 10^{-10}$ m^2/s). It is important to note that, since c is itself a function of r_H, the typical protocol for quantitative analysis consists in using the SE to estimate the cr_H product from experimental D_t:

$$cr_H = \frac{k_B T}{f_s \pi \eta D_t} \tag{3}$$

and then using a modification of Eq. 2, obtained by multiplying both terms by r_H:

$$cr_H = \frac{6}{1 + 0.695\left(\frac{r_{solv.}}{r_H}\right)^{2.234}} \times r_H \tag{4}$$

to build a solvent specific cr_H vs. r_H graph. This graph provides a numerical solution to the problem, allowing to extract r_H for any cr_H value derived from experimental data (Equation (3)).[11]

In dichloromethane-d_2, the estimated hydrodynamic volume (V_H) of BF_4^- is 57 ± 8 Å^3, consistent with the expected value, considering that the van der Waals volume (V_{vdW}) of this anion is 42 Å^3 (Table 1, entries 1).[26] The observed V_H/V_{vdW} ratio of ~1.35 aligns with typical observations in organic solvents ($V_{vdW} = V_H$ only for perfectly spherical and dense particles).[11,26]

In contrast, applying the same method in water yields a V_H of 30 ± 4 Å^3 (Table 1, entry 3), a value that is clearly underestimated. Even less accurate estimates are obtained when applying the SE equation in its original formulation, both under "stick boundary" ($c = 6$) or "slip boundary" conditions ($c = 4$; Table 1, entries 4–5).[11,16]

Table 1. Comparison between V_{vdW} and V_H (estimated under different approximations for c and in various solvents) of BF_4^-, exemplifying the difficulty in analyzing aqueous solutions.

Entry	Solvent	Approximation for c[a]	V_H (Å^3)[b]
1	*van der Waals volume*		42
2	CD_2Cl_2	Chen and Chen (Equation (4))	57 ± 8
3	D_2O	Chen and Chen (Equation (4))	30 ± 4
4	D_2O	"Stick boundary" (c = 6)	8 ± 1
5	D_2O	"Slip boundary" (c = 4)	26 ± 4

[a] $r_{CD2Cl2} = 2.49$ Å; $r_{D2O} = 1.93$ Å. [b] Determined by diffusion NMR at 298 K; 15% experimental uncertainty assumed on V_H.[46]

Development of a Water-Specific Equation for *c*

Evidently, the problem in water lies in the fact that Equation (4) does not fit the cr_H vs. r_H profile accurately, leading to erroneous estimates of r_H (*vide infra*). To address this limitation, we derived an empirical $c = f\,(r_{solv}, r_H)$ expression specific for water, analyzing the D_t of a variety of known molecules varying in polarity, charge, hydrophilicity, acid-base properties, and hydrogen-bond capabilities (Figure 1). These molecules, encompassing gases, small organic compounds, amino acids, oligopeptides, saccharides, salts, and coordination complexes, provides a comprehensive dataset for constructing an experimental cr_H vs. r_H plot over a wide range of molecular volumes (from ~10^1–10^3 Å^3).

The cr_H values were derived from D_t (Equation (3)), which were measured experimentally by DOSY NMR or taken from the literature. The partial molar radius at infinite dilution (r_M) was used as a reliable proxy for the otherwise unknown r_H.[11] For most molecules, particularly smaller ones, f_s can be approximated to 1; f_s significantly deviates from that of a perfect sphere only when the ratio between the two semiaxis of elongated molecules, modelled as ellipsoids, is higher than 3.[11–12] The non-spherical shape of the other species, *e.g.* oligopeptides and oligosaccharides, was considered according to the model proposed by Perrin.[17]

The resulting cr_H vs. r_M plot revealed significant scatter in the data points, leading to a poorly defined trend (Figure 2a). A closer inspection reveals that the largest deviations are observed for molecules containing polar functionalities capable of forming strong hydrogen bonds, such as carboxylic (–COOH), hydroxy (–OH), amino (–NH or –NH_2) and imino (=N–) groups. Representative examples are highlighted in Figure 2b.

Refinement to Account for Hydrogen Bonding

To account for these deviations, we introduced a correction to r_M, adding the volume of one water molecule ($V_{H_2O} = 30.0$ Å^3)[47] for each functional group forming strong hydrogen bonds. This "corrected" r_M (r^*_M) led to a more consistent alignment of data points, resulting in a significantly improved cr_H vs. r^*_M plot (Figure 2c and d).

This correction is justified by the fact that solute-solvent interactions, particularly hydrogen bonding, significantly influence hydrodynamic dimensions.[48–52] Water molecules strongly interacting with polar solutes diffuse with the analyte, contributing to its V_H:[53–57] this "effective" volume is the one actually determining diffusivities and, consequently, important properties for *e.g.* electrochemical,[57] biological[49,52] and industrial applications.[50] The correction is particularly pronounced for polar groups capable of forming strong hydrogen bonds like

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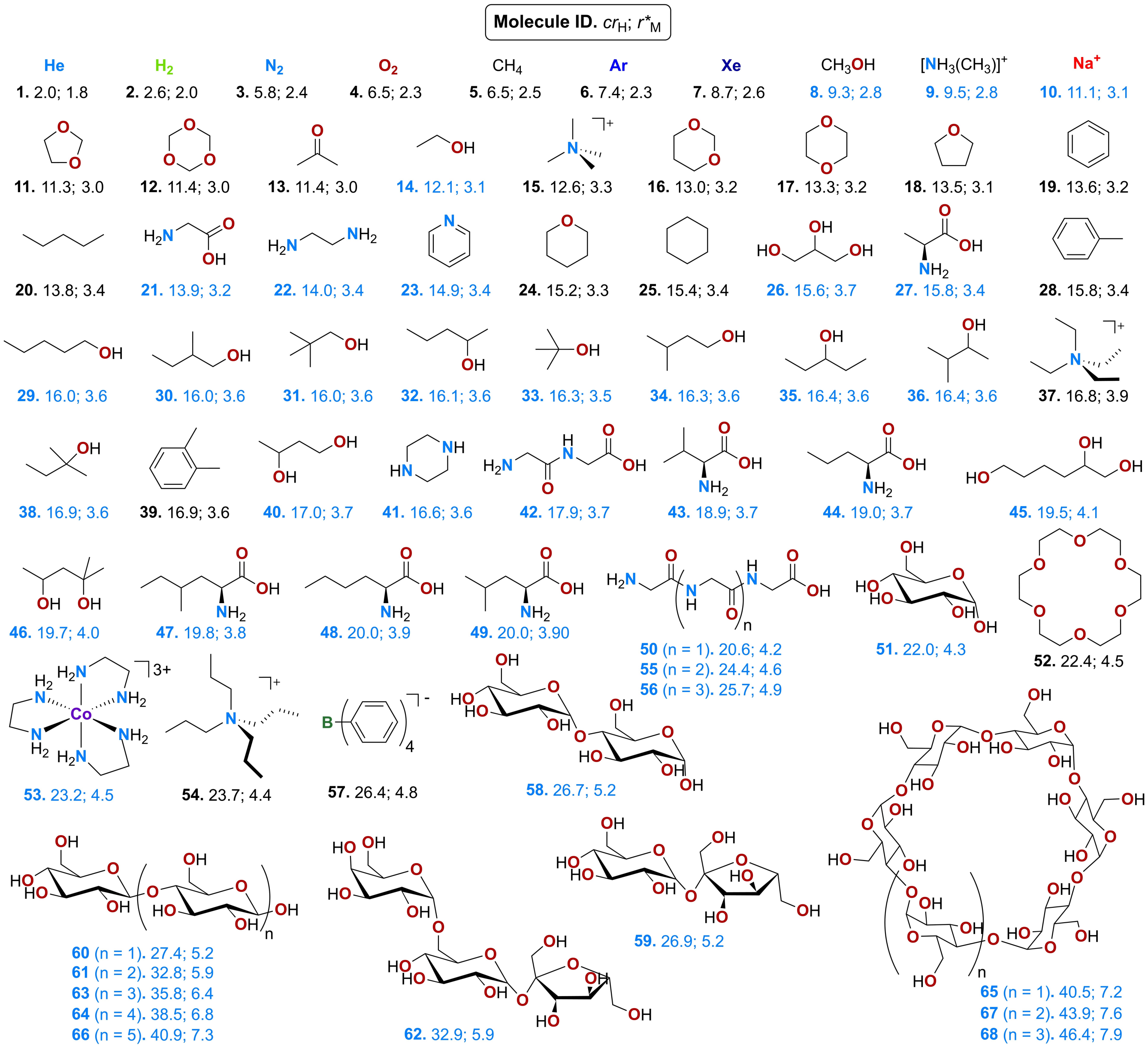

Figure 1. Calibration dataset of compounds used to build-up the plots reported in Figure 2. Each molecule is reported with its ID, cr_H (Å, estimated from experimental D_t) and r^*_M (Å). Datapoints are reported according to increasing cr_H. Species establishing strong hydrogen bonding with water are highlighted in blue. Experimental data were measured herein or derived from the literature.[60–91] See Table S1 in the Supporting Information for additional details.

the abovementioned carboxylic, hydroxy, amino and imino functionalities. Conversely polar groups establishing weaker interactions with water, like ethers and carbonyls,[58–59] do not require this correction, just like apolar species (Figure 2b and d).

Despite this refinement of the data set, existing methods provided unsatisfactory fits to the cr_H vs. r^*_M plot for small and medium-sized molecules (grey lines in Figure 2c). Specifically, the Chen and Chen equation underestimates the size especially of the smallest molecules, reflecting the inadequacies highlighted earlier with BF_4^-. Also attempts to fit experimental data with one of the most recently proposed semiempirical correlations between D_t and molecular size of the solute, expressed in terms of molecular weight (MW),[15] provided unsatisfactory results (~9% average error on D_t; see Table S2).

To address these shortcomings, we empirically refined the parameters of Equation (4). Recognizing the central role of hydrogen bonding for solutions in water, we focused on r_{H_2O} and the exponent of the r_{H_2O}/r_H ratio. The rationale behind this is that the former parameter is likely affected by the hydrogen bonds among water molecules, just like discussed above for solute-solvent interactions. The second parameter is instead the one mostly determining the dependence of c on the r_{H_2O}/r_H ratio and, consequently, the shape of the cr_H vs. r^*_M profile.

The best fit was obtained with the following equation:

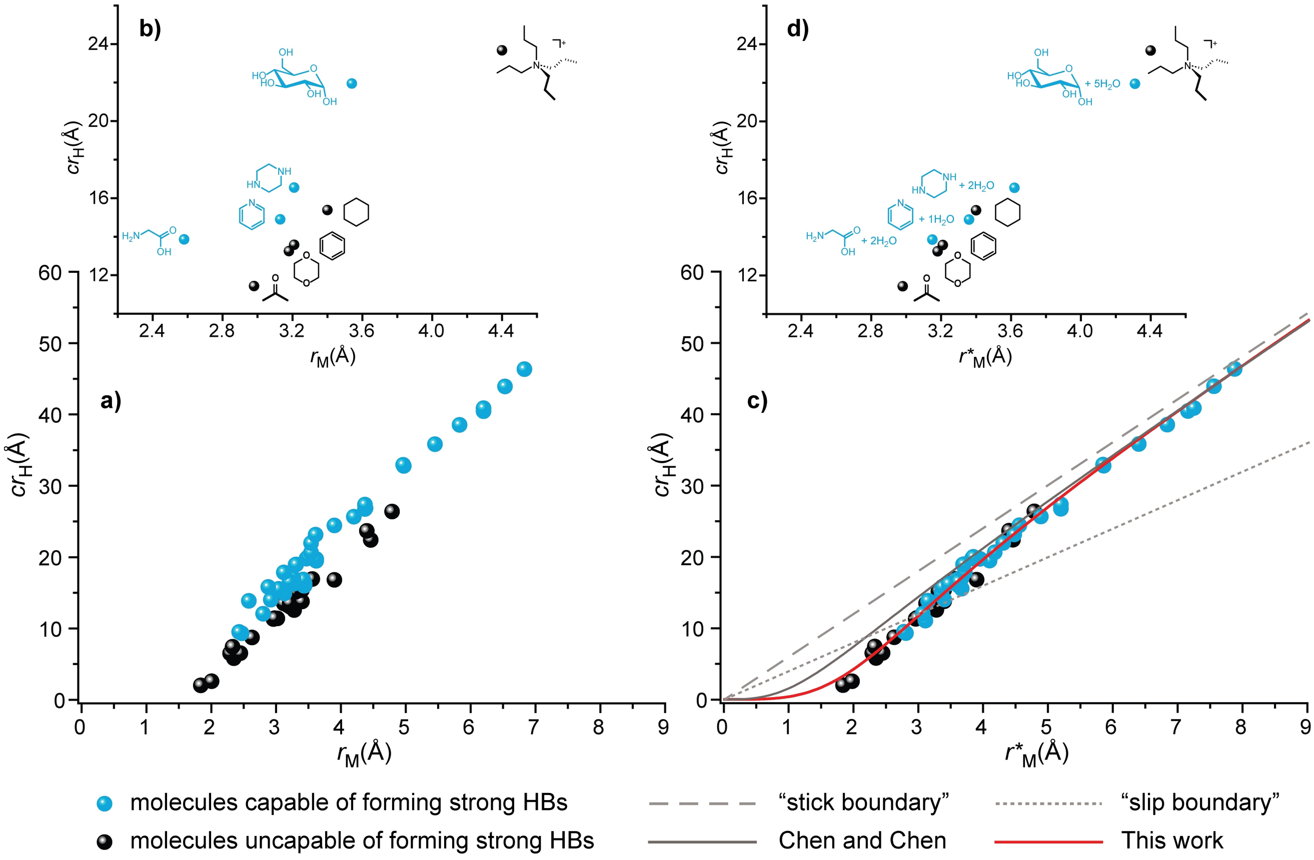

Figure 2. Comparison between a) cr_H vs. r_M and c) cr_H vs. r^*_M plots for the studied molecules. Inserts show a selection of the datapoints with b) uncorrected r_M and d) corrected r^*_M. Data fitting of the cr_H vs. r^*_M plots using established equations (grey lines) vs. herein developed equation (Equation (5), red line; $R^2 = 0.99$) is reported. Datapoint for Na^+ ($cr_H = 13.3$ Å vs. $r_M = -1.38$ Å) is omitted for clarity in the cr_H vs. r_M graph. Species establishing strong hydrogen bonds (HBs) with water are highlighted in blue.

$$cr_H = \frac{6}{1 + 0.695\left(\frac{2.78}{r_H}\right)^{3.044}} \times r_H \tag{5}$$

in which r_{H_2O} is set to 2.78 Å (*i.e.* approximately three water molecules) and the exponent is slightly increased to 3.044. This refined equation provided an excellent fit to the experimental data, with a R^2 value of 0.99 (red curve, Figure 2c) and an a priori average maximum error in D_t and r_H of only ~5% and 3%, respectively (Supporting Information). Therefore, it can be conservatively assumed that the error in estimating hydrodynamic dimensions from experimental D_t is comparable to the Chen and Chen method for organic solvents, namely 5%for r_H and 15% for V_H.[11,46]

With this refined equation, it is also possible to derive the c values for all the studied molecules (Figure S3). Consistently with expectations, this parameter varies significantly across the dataset, approaching the ideal value of 6 ("stick boundary" conditions) only for the largest species. For some medium- to small-sized molecules, c can reasonably be approximated to 4 ("slip boundary" conditions), while it drops to ~1 for the smallest species like He and H_2.

Validation and Applicability of the Approach

The effectiveness of the proposed equation was tested on several representative case studies. The first test-set is composed of seven representative small molecules (Figure 3a and Table S3), which – as highlighted above – represent the most challenging class of analytes. They include the previously discussed tetrafluoroborate anion, an apolar alkyl benzene, and five neutral organic molecules capable of forming strong hydrogen bonds with up to three water molecules.

The estimated V_H values using Equation (5) show excellent agreement with expected ones (*i.e.* V^*_M; Figure 3a), validating the robustness of the proposed equation. The method can capture the formation of hydrogen bonds with water, as demonstrated for instance by serine, which appears larger than 1-methyl-2-pyrrolidone despite having a smaller uncorrected V_M (Table S3).

The second set of test molecules features two organometallic complexes with 2,2′-dipyridylketone (dpk) ligands, namely **[(Cym)Ru(κ^3-dpk-OH]$^+$**[94] and the novel **[(Cp*)Ir(κ^3-dpk-OH]$^+$** (Figure 3b and Supporting Information). These representative complexes are interesting in two respects. First, their V_M are unavailable in the literature and difficult to measure due to the need for large amounts of such expensive noble metal

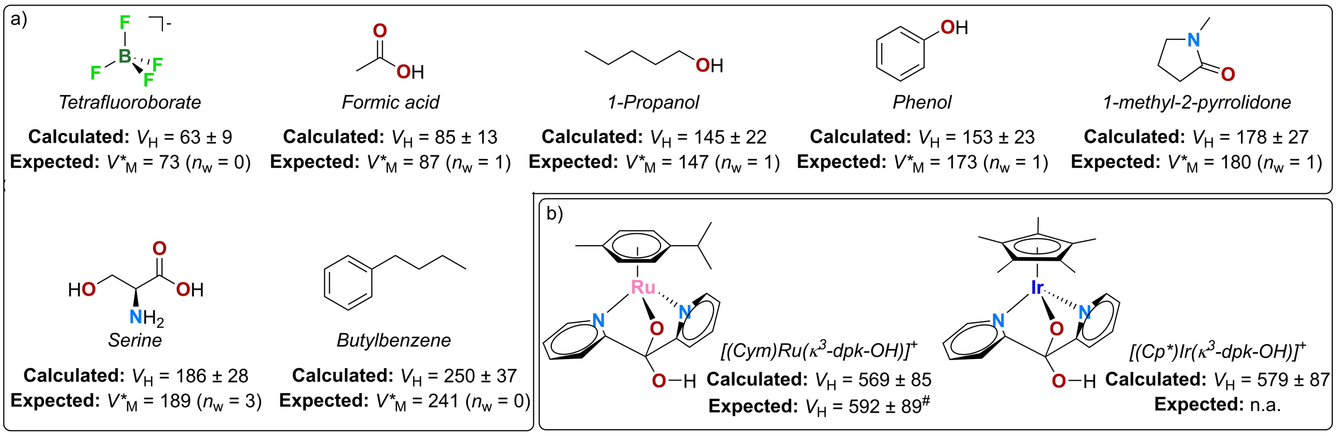

Figure 3. Molecules used as test-set for validating Equation (5). The V_H, determined experimentally in water at 298 K with Equation (5), are reported along with reference values taken from the literature.[47,65–67,92–93] The number of associated water molecules (n_w) is highlighted. # Determined experimentally for the methoxy analogue, [(Cym)Ru(κ^3-dpk-OMe]$^+$, in dichloromethane-d_2 at 298 K.[94] All volumes are reported in Å^3. See Table S3 in the Supporting Information for additional details.

compounds. Therefore, they represent ideal candidates for DOSY NMR measurements, which can provide molecular size information from only milligram-scale samples. Second, the presence of heavy metal atoms makes their molecular density appreciably different from that of organic molecules, complicating the use of molecular weight-based size estimation methods.[12,95] In fact, applying an advanced MW estimation equation[15] to **[(Cym)Ru(κ^3-dpk-OH]$^+$** ($D_t = 4.26\times10^{-10}$ m^2/s)[94] and **[(Cp*)Ir(κ^3-dpk-OH]$^+$** ($D_t = 4.21\times10^{-10}$ m^2/s) results in 20–32% underestimation of molecular weight (346 Da and 356 Da, respectively) compared to the actual values (436 Da and 529 Da, respectively).

In contrast, Equation (5) provides two similar V_H of approximatively 569 Å^3 and 579 Å^3 for **[(Cym)Ru(κ^3-dpk-OH]$^+$** and **[(Cp*)Ir(κ^3-dpk-OH]$^+$**, respectively. The similarity of these values is consistent with the nearly isosteric nature of these two compounds. Furthermore, they are comparable to the V_H of the methoxy analogue of the Ru complex, **[(Cym)Ru(κ^3-dpk-OMe]$^+$** ($D_t = 11.4\times10^{-10}$ m^{2}s; $V_H = 592$ Å^3),[94] estimated with the Chen and Chen equation in dichloromethane-d_2 (*i.e.* within its range of validity). These results demonstrate the broad applicability and high accuracy of our method, making it a valuable tool for determining hydrodynamic dimensions in a wide range of unknown analytes.

Conclusions

A semiempirical equation (Equation (5)) has been developed to enable the accurate estimation of hydrodynamic dimensions in water also for small and medium-sized molecules, addressing one of the main limitations of existing methods for the quantitative analysis of D_t. This equation, derived from a comprehensive analysis of nearly 70 molecules with varying chemical properties and size, accounts for the structured nature of water and its ability to form strong hydrogen bonds.

This result was achieved by empirically refining the parameters of the equation proposed by Chen and Chen[23] for the correction factor *c* in the modified SE equation. Specifically, the hydrodynamic radius of the solvent is adjusted from that of a single water molecule ($r_{H_2O} = 1.93$ Å, $V_{H_2O} = 30.0$ Å^3) to approximately three water molecules ($r_{H_2O} = 2.78$ Å, $V_{H_2O} = 90.0$ Å^3), along with modifications to the exponent of the r_{H_2O}/r_H ratio. Furthermore, the r_H of the solute is adjusted to include one associated water molecule for each polar functional group capable of forming strong hydrogen bonds.

This new approach facilitates more precise molecular characterization in water, one of the most challenging and ubiquitous solvents. Given that measuring D_t is becoming a routine task in modern chemical laboratories, the equation presented here offers a powerful means for advancing the quantitative interpretation of these information-rich parameters, with broad applicability to chemical and biological systems. To name but a few, potential applications include the study of host-guest interactions, biomolecules folding and aggregation, aggregation of organometallic complexes and formation of hydrogen bonds in water.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: DOSY NMR · Stokes-einstein equation · Hydrodynamic volume · Hydrogen bonding · Aqueous systems

[1] E. N. Saw, N. Blanc, K. Kanokkanchana, K. Tschulik, *Electrochim. Acta* **2018**, *282*, 317–323.
[2] C. T. Culbertson, S. C. Jacobson, J. Michael Ramsey, *Talanta* **2002**, *56*, 365–373.
[3] R. Liu, X. Gao, J. Adams, W. Oppermann, *Macromolecules* **2005**, *38*, 8845–8849.
[4] G. Pagès, V. Gilard, R. Martino, M. Malet-Martino, *Analyst* **2017**, *142*, 3771–3796.
[5] C. S. Johnson, *Prog. Nucl. Magn. Reson. Spectrosc.* **1999**, *34*, 203–256.
[6] D. Li, I. Keresztes, R. Hopson, P. G. Williard, *Acc. Chem. Res.* **2009**, *42*, 270–280.
[7] C. Lhoste, B. Lorandel, C. Praud, A. Marchand, R. Mishra, A. Dey, A. Bernard, J.-N. Dumez, P. Giraudeau, *Prog. Nucl. Magn. Reson. Spectrosc.* **2022**, *130–131*, 1–46.
[8] W. Li, H. Chung, C. Daeffler, J. A. Johnson, R. H. Grubbs, *Macromolecules* **2012**, *45*, 9595–9603.
[9] Y. Cohen, L. Avram, L. Frish, *Angew. Chem. Int. Ed.* **2005**, *44*, 520–554.
[10] G. Bellachioma, G. Ciancaleoni, C. Zuccaccia, D. Zuccaccia, A. Macchioni, *Coord. Chem. Rev.* **2008**, *252*, 2224–2238.
[11] A. Macchioni, G. Ciancaleoni, C. Zuccaccia, D. Zuccaccia, *Chem. Soc. Rev.* **2008**, *37*, 479–489.
[12] F. Zaccaria, C. Zuccaccia, R. Cipullo, A. Macchioni, *Chem. - Eur. J.* **2019**, *25*, 9930–9937.
[13] R. Evans, *Prog. Nucl. Magn. Reson. Spectrosc.* **2020**, *117*, 33–69.
[14] R. Evans, Z. Deng, A. K. Rogerson, A. S. McLachlan, J. J. Richards, M. Nilsson, G. A. Morris, *Angew. Chem. Int. Ed.* **2013**, *52*, 3199–3202.
[15] R. Evans, G. Dal Poggetto, M. Nilsson, G. A. Morris, *Anal. Chem.* **2018**, *90*, 3987–3994.
[16] J. T. Edward, *J. Chem. Educ.* **1970**, *47*, 261.
[17] F. Perrin, *J. Phys. Radium.* **1936**, *7*, 1.
[18] P. H. Elworthy, *J. Chem. Soc.* **1962**, 3718–3723.
[19] L. Wang, R. Liu, J. Gu, B. Song, H. Wang, X. Jiang, K. Zhang, X. Han, X. Q. Hao, S. Bai, M. Wang, X. Li, B. Xu, X. Li, *J. Am. Chem. Soc.* **2018**, *140*, 14087–14096.
[20] B. M. Schulze, D. L. Watkins, J. Zhang, I. Ghiviriga, R. K. Castellano, *Org. Biomol. Chem.* **2014**, *12*, 7932–7936.
[21] A. Gierer, K. Wirtz, *Zeitschrift fur Naturforsch. - Sect. A J. Phys. Sci.* **1953**, *8*, 532–538.
[22] A. Spernol, K. Wirtz, *Zeitschrift fur Naturforsch. - Sect. A J. Phys. Sci.* **1953**, *8*, 522–532.
[23] H. C. Chen, S. H. Chen, *J. Phys. Chem.* **1984**, *88*, 5118–5121.
[24] D. Beaudoin, F. Rominger, M. Mastalerz, *Angew. Chem. Int. Ed.* **2016**, *55*, 15599–15603.
[25] M.-E. Moret, P. Chen, *J. Am. Chem. Soc.* **2009**, *131*, 5675–5690.
[26] D. Zuccaccia, G. Bellachioma, G. Cardaci, G. Ciancaleoni, C. Zuccaccia, E. Clot, A. Macchioni, *Organometallics* **2007**, *26*, 3930–3946.
[27] C. Lindon John, Liu Maili, K. Nicholson Jeremy, *Rev. Anal. Chem.* **1999**, *18*, 23–66.
[28] E. F. Dudás, A. Bodor, *Anal. Chem.* **2019**, *91*, 4929–4933.
[29] B. Tang, K. Chong, W. Massefski, R. Evans, *J. Phys. Chem. B* **2022**, *126*, 5887–5895.
[30] J. Lee, S. H. Park, S. Cavagnero, J. H. Lee, *Anal. Chem.* **2020**, *92*, 5073–5081.
[31] G. Kleks, D. C. Holland, J. Porter, A. R. Carroll, *Chem. Sci.* **2021**, *12*, 10930–10943.
[32] W. Ge, J. H. Zhang, C. M. Pedersen, T. Zhao, F. Yue, C. Chen, P. Wang, Y. Wang, Y. Qiao, *ACS Sustain. Chem. Eng.* **2016**, *4*, 1193–1200.
[33] L. Wang, W. Amelung, S. Willbold, *Environ. Sci. Technol.* **2017**, *51*, 13256–13264.
[34] G. Peat, P. J. Boaler, C. L. Dickson, G. C. Lloyd-Jones, D. Uhrín, *Nat. Commun.* **2023**, *14*, 4410.
[35] Z. Laughrey, B. C. Gibb, *Chem. Soc. Rev.* **2011**, *40*, 363–386.
[36] N. Watfa, D. Melgar, M. Haouas, F. Taulelle, A. Hijazi, D. Naoufal, J. B. Avalos, S. Floquet, C. Bo, E. Cadot, *J. Am. Chem. Soc.* **2015**, *137*, 5845–5851.
[37] M. D. Pluth, B. E. F. Tiedemann, H. van Halbeek, R. Nunlist, K. N. Raymond, *Inorg. Chem.* **2008**, *47*, 1411–1413.
[38] A. Gafni, Y. Cohen, *J. Org. Chem.* **1997**, *62*, 120–125.
[39] E. Aguiló, A. J. Moro, R. Gavara, I. Alfonso, Y. Pérez, F. Zaccaria, C. F. Guerra, M. Malfois, C. Baucells, M. Ferrer, J. C. Lima, L. Rodríguez, *Inorg. Chem.* **2018**, *57*, 1017–1028.
[40] I. Corbucci, A. Petronilho, H. Müller-Bunz, L. Rocchigiani, M. Albrecht, A. Macchioni, *ACS Catal.* **2015**, *5*, 2714–2718.
[41] A. Savini, P. Belanzoni, G. Bellachioma, C. Zuccaccia, D. Zuccaccia, A. Macchioni, *Green Chem.* **2011**, *13*, 3360–3374.
[42] G. Menendez Rodriguez, F. Zaccaria, L. Tensi, C. Zuccaccia, P. Belanzoni, A. Macchioni, *Chem. - Eur. J.* **2021**, *27*, 2050–2064.
[43] G. M. Rodriguez, F. Zaccaria, S. Van Dijk, C. Zuccaccia, A. Macchioni, *Organometallics* **2021**, *40*, 3445–3453.
[44] R. Piazza, V. Degiorgio, *J. Phys. Condens. Matter* **1993**, *5*, B173–B182.
[45] G. M. Kontogeorgis, A. Holster, N. Kottaki, E. Tsochantaris, F. Topsøe, J. Poulsen, M. Bache, X. Liang, N. S. Blom, J. Kronholm, *Chem. Thermodyn. Therm. Anal.* **2022**, *6*, 100053.
[46] D. Zuccaccia, A. Macchioni, *Organometallics* **2005**, *24*, 3476–3486.
[47] S. D. Hamann, S. C. Lim, *Aust. J. Chem.* **1954**, *7*, 329–334.
[48] T. C. Chan, C. H. C. C. Chan, W. Y. Tang, N. W. Chang, *J. Phys. Chem. B* **2018**, *122*, 9236–9249.
[49] T. C. Chan, Y. Y. Lui, T. L. H. Lam, C. H. C. Chan, *Chem. Eng. Sci.* **2021**, *233*, 116205.
[50] R. Sinibaldi, C. Casieri, S. Melchionna, G. Onori, A. L. Segre, S. Viel, L. Mannina, F. De Luca, *J. Phys. Chem. B* **2006**, *110*, 8885–8892.
[51] C. M. Silva, H. Liu, E. A. Macedo, *Chem. Eng. Sci.* **1998**, *53*, 2423–2429.
[52] A. L. Magalhães, F. A. Da Silva, C. M. Silva, *J. Supercrit. Fluids* **2013**, *74*, 89–104.
[53] G. S. Kapur, E. J. Cabrita, S. Berger, *Tetrahedron Lett.* **2000**, *41*, 7181–7185.
[54] E. J. Cabrita, S. Berger, *Magn. Reson. Chem.* **2001**, *39*, S142–S148.
[55] H.-H. Limbach, *Magn. Reson. Chem.* **2001**, *39*, S1–S2.
[56] O. Mayzel, A. Gafni, Y. Cohen, *Chem. Commun.* **1996**, 911–912.
[57] K. Hayamizu, Y. Chiba, T. Haishi, *RSC Adv.* **2021**, *11*, 20252–20257.
[58] L. N. Ferguson, *J. Am. Chem. Soc.* **1955**, *77*, 5288–5289.
[59] J. P. M. Lommerse, S. L. Price, R. Taylor, *J. Comput. Chem.* **1997**, *18*, 757–774.
[60] D. L. Wise, G. Houghton, *Chem. Eng. Sci.* **1966**, *21*, 999–1010.
[61] W. M. Haynes, *CRC Handbook of Chemistry and Physics*, 95th Ed., CRC Press, Boca Raton **2014**.
[62] H. Ping, D. M. Bartels, *J. Phys. Chem.* **1996**, *100*, 5597–5602.
[63] R. E. Smith, E. T. Friess, M. F. Morales, *J. Phys. Chem.* **1955**, *59*, 382–383.
[64] L. Lepori, P. Gianni, *J. Solution Chem.* **2000**, *29*, 405–447.
[65] J. Šedlbauer, *Standard Molar Volumes of Aqueous Organics: "Database on the Infinite Dilution Partial Molar Volumes of Aqueous Organic Nonelectrolytes and Electrolytes"*. https://www.kch.tul.cz/sedlbauer/vorg.nb.
[66] F. J. Millero, *Chem. Rev.* **1971**, *71*, 147–176.
[67] M. J. M. te Riele, E. D. Snijder, W. P. M. van Swaaij, *J. Chem. Eng. Data* **1995**, *40*, 34–36.
[68] S. B. Rempe, L. R. Pratt, *Fluid Phase Equilib.* **2001**, *183–184*, 121–132.
[69] H. G. Hertz, *Water: A Comprehensive Treatise*, Vol. *3* (Ed: F. Franks), Plenum Press, New York **1973**.
[70] F. J. Millero, *Water and Aqueous Solutions*, Ch. 13 (Ed: R. H. Horne), Wiley, New York **1972**.
[71] D. G. Leaist, K. MacEwan, A. Stefan, M. Zamari, *J. Chem. Eng. Data* **2000**, *45*, 815–818.
[72] I. Cibulka, C. Alexiou, *J. Chem. Thermodyn.* **2010**, *42*, 274–285.
[73] T. Funazukuri, M. Nishio, *J. Chem. Eng. Data* **1999**, *44*, 73–76.
[74] D. Fenclová, S. Perez-Casas, M. Costas, V. Dohnal, *J. Chem. Eng. Data* **2004**, *49*, 1833–1838.
[75] T. Gabler, A. Paschke, G. Schüürmann, *J. Chem. Eng. Data* **1996**, *41*, 33–36.
[76] T. Tominaga, S. Yamamoto, J.-I. Takanaka, *J. Che. Soc. Farady Trans 1* **1984**, *80*, 941.
[77] W. S. Price, O. Söderman, *J. Phys. Chem. A* **2000**, *104*, 5892–5894.

[78] T. Umecky, T. Kuga, T. Funazukuri, *J. Chem. Eng. Data* **2006**, *51*, 1705–1710.
[79] H. Zhao, *Biophys. Chem.* **2006**, *122*, 157–183.
[80] R. Niesner, A. Heintz, *J. Chem. Eng. Data* **2000**, *45*, 1121.
[81] V. Majer, S. Degrange, J. Sedlbauer, *Fluid Phase Equilib.* **1999**, *158–160*, 419–428.
[82] T. Tominaga, S. Matsumoto, *J. Chem. Eng. Data* **1990**, *35*, 45–47.
[83] P. Hynčica, L. Hnědkovský, I. Cibulka, *J. Chem. Thermodyn.* **2006**, *38*, 418–426.
[84] P. W. J. Derks, E. S. Hamborg, J. A. Hogendoorn, J. P. M. Niederer, G. F. Versteeg, *J. Chem. Eng. Data* **2008**, *53*, 1179–1185.
[85] M. Ishimura, H. Uedaira, *Bull. Chem. Soc. Jpn.* **1990**, *63*, 1–5.
[86] S. Sirimulla, M. Lerma, W. C. Herndon, *J. Chem. Inf. Model.* **2010**, *50*, 194–204.
[87] F. Capuano, G. Mangiapia, G. D'Errico, R. Sartorio, *J. Solution Chem.* **2006**, *35*, 1525–1535.
[88] F. Kawaizumi, F. Nakao, H. Nomura, *J. Solution Chem.* **1985**, *14*, 687–697.
[89] C. Monteiro, C. Hervé du Penhoat, *J. Phys. Chem. A* **2001**, *105*, 9827–9833.
[90] M. Ihnat, D. A. I. Goring, *Can. J. Chem.* **1967**, *45*, 2353–2361.
[91] A. C. F. Ribeiro, A. J. M. Valente, C. I. A. V Santos, P. M. R. A. Prazeres, V. M. M. Lobo, H. D. Burrows, M. A. Esteso, A. M. T. D. P. V. Cabral, F. J. B. Veiga, *J. Chem. Eng. Data* **2007**, *52*, 586–590.
[92] O. Großmann, D. Bellaire, N. Hayer, F. Jirasek, H. Hasse, *Digit. Discov.* **2022**, *1*, 886–897.
[93] T. Tominaga, S. Matsumoto, T. Ishii, *J. Phys. Chem.* **1986**, *90*, 139–143.
[94] D. Zuccaccia, E. Foresti, S. Pettirossi, P. Sabatino, C. Zuccaccia, A. Macchioni, *Organometallics* **2007**, *26*, 6099–6105.
[95] R. Neufeld, D. Stalke, *Chem. Sci.* **2015**, *6*, 3354–3364.
[96] A. Jerschow, N. Müller, *J. Magn. Reson.* **1997**, *125*, 372–375.
[97] E. O. Stejskal, J. E. Tanner, *J. Chem. Phys.* **1965**, *42*, 288–292.
[98] R. Mills, *J. Phys. Chem.* **1973**, *77*, 685–688.
[99] E. M. M. Del Valle, *Process Biochem.* **2004**, *39*, 1033–1046.
[100] M. Ihnat, D. A. I. Goring, *Can. J. Chem.* **1967**, *45*, 2363–2367.

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